Question 1

a) because we observe a molecule only conditional on $K_i > 0$ the relevant distribution is the poisson conditioned on being > 0:

$$P_{obs}(K_i; \lambda) = P(K_i \mid K_i > 0) = \frac{P(K_i)}{P(K_i > 0)}$$

$$= \frac{\lambda^{K_i} e^{-\lambda}}{K_i! (1 - e^{-\lambda})} \quad \text{s.t} \quad K_i \geq 1$$

dividing by $(1 - e^{-\lambda})$ makes the probabilities sum to 1 again after dropping the $K_i = 0$ case.

b) let M_e be the number of observed molecules (the num of factors of the product). the likelihood is

$$L(\lambda) = \prod_{i: K_i > 0} \frac{\lambda^{K_i} e^{-\lambda}}{K_i! (1 - e^{-\lambda})}$$

$$\Rightarrow \ell(\lambda) = \sum_{i: K_i > 0} \left[K_i \log \lambda - \lambda - \log(K_i!) - \log(1 - e^{-\lambda}) \right]$$

now use that $\sum_{i: K_i > 0} K_i = \sum_i K_i = M$

(molecules with $K_i = 0$ doesn't change the sum)

$$\rightarrow \ell(\lambda) = M \log \lambda - M_u \lambda - M_u \log(1-e^{-\lambda}) - \underbrace{\sum_i \log(K_i!)}_{\text{doesn't depend on } \lambda}$$

the data affects the likelihood only through two nums: $M = \sum_i K_i$ (total reads) and M_u (Num of unique observed molecules). these are the sussicient statistics. and they're exactly the two quantities you can read directly off the data.

c) derivative of $-M_u \log(1-e^{-\lambda})$ is $-M_u \frac{e^{-\lambda}}{1-e^{-\lambda}}$ so:

$$\frac{d\ell}{d\lambda} = \frac{M}{\lambda} - M_u - M_u \frac{e^{-\lambda}}{1-e^{\lambda}} = 0$$

Combine the two M_u terms:

$$M_u + M_u \frac{e^{-\lambda}}{1-e^{-\lambda}} = M_u \frac{(1-e^{-\lambda}) + e^{-\lambda}}{1-e^{-\lambda}} =$$

$$= \frac{M_u}{1-e^{-\lambda}}$$

=)

$$\frac{M}{\lambda} = \frac{M_u}{1-e^{-\lambda}} \iff \frac{M_u}{M} = \frac{1-e^{-\lambda}}{\lambda}$$

⇒ maximizing the likelihood gives the same equation as the method from class.

$\frac{M_u}{M} \approx \frac{1-e^{-\lambda}}{\lambda}$. So both methods produce the same estimates of λ.

d)

$$E[M_u] = N(1-e^{\lambda}).$$

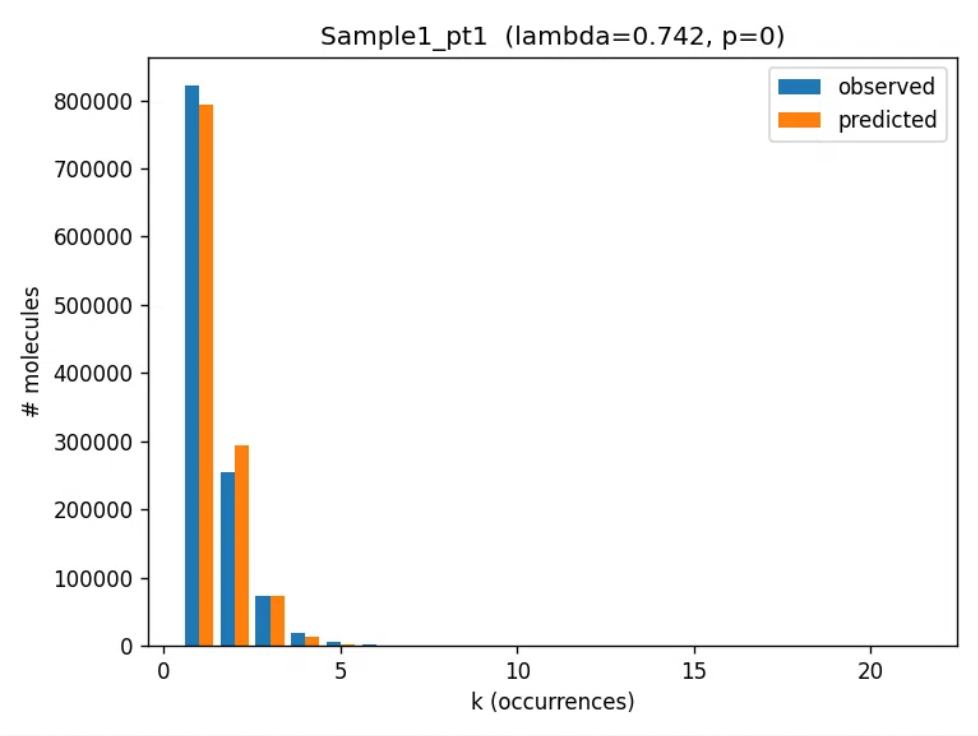
we plug $\hat{\lambda}$ and M_u:

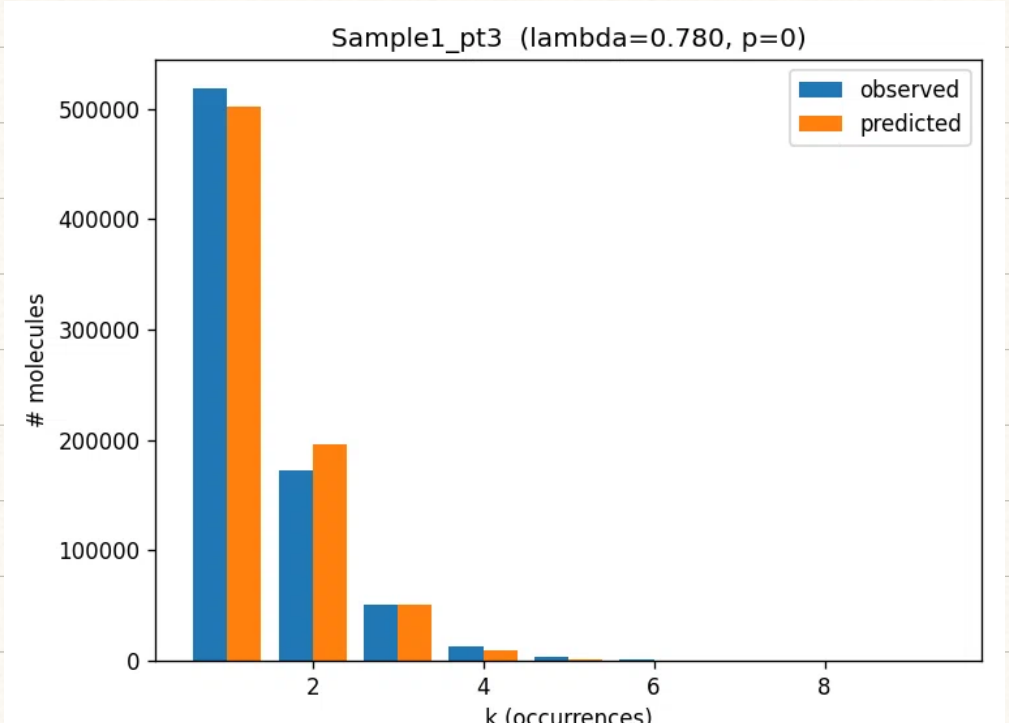
$$\hat{N} = \frac{M_u}{1-e^{-\hat{\lambda}}}.$$

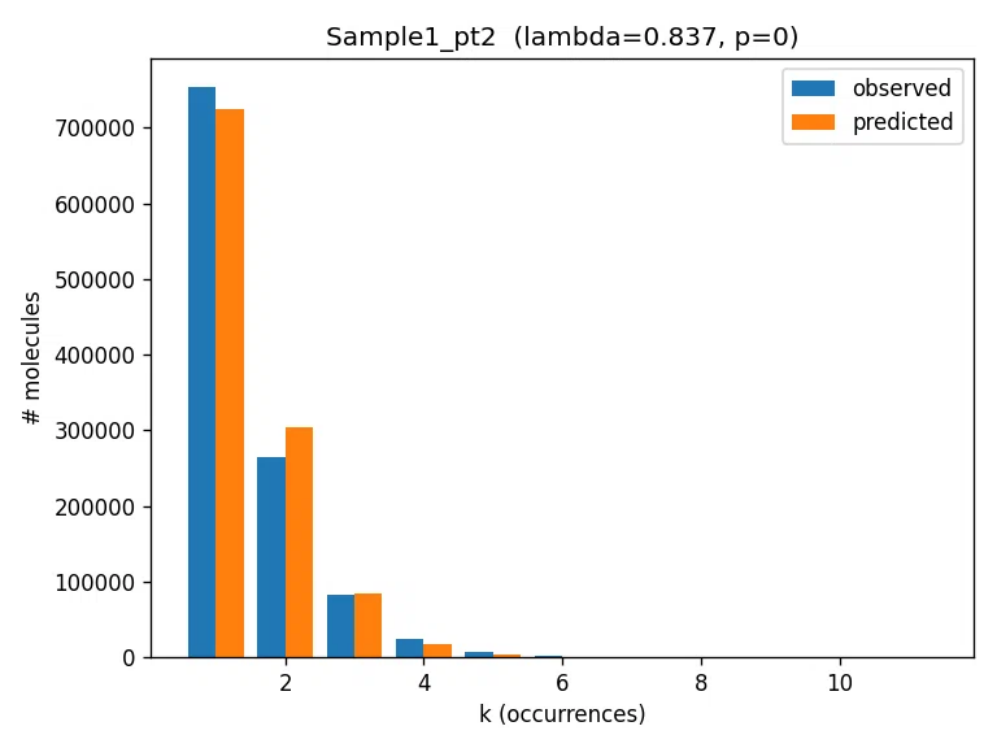
Question 2

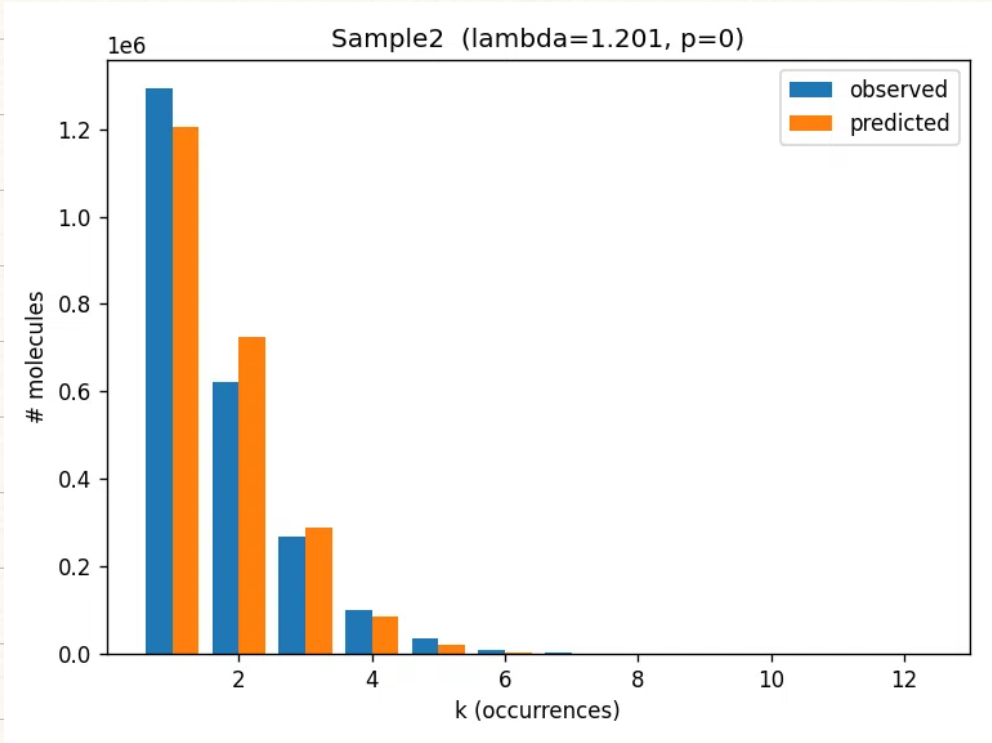
a) the 2 methods give identical estimates, every line printed, and the $\hat{\lambda}$ values are (0.742, 0.837, 0.780, 1.201) are the same numbers from both. class method == MLE

b)









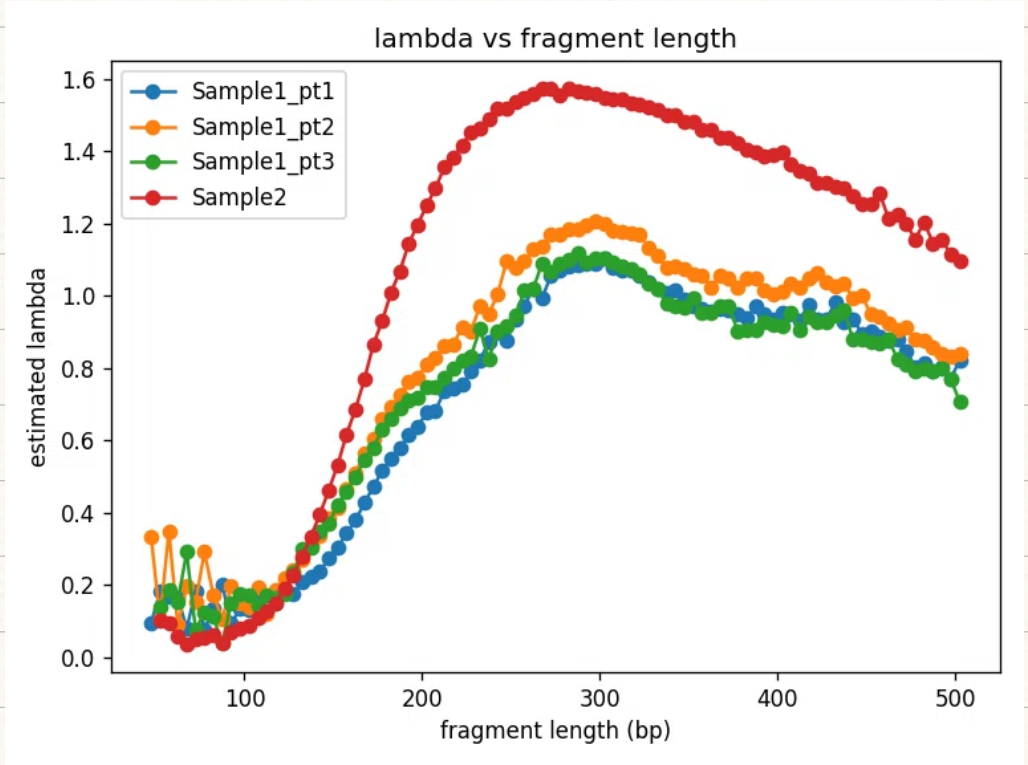
in all 4 samples the fitted Z model misses the data the same way, it underestimates the num of molecules seen once, overestimates those who were seen twice, and underestimates the high count tail.

the counts are more spread out than a single

poisson permits.
i tested with Chi squared goodness of fit and every sample give $p \approx 0$, so the single λ model is not accepted.
this happens since each file pools fragments of many lengths and since λ is dependent on that, the pooled counts are mix of poissons which is wider than one poisson

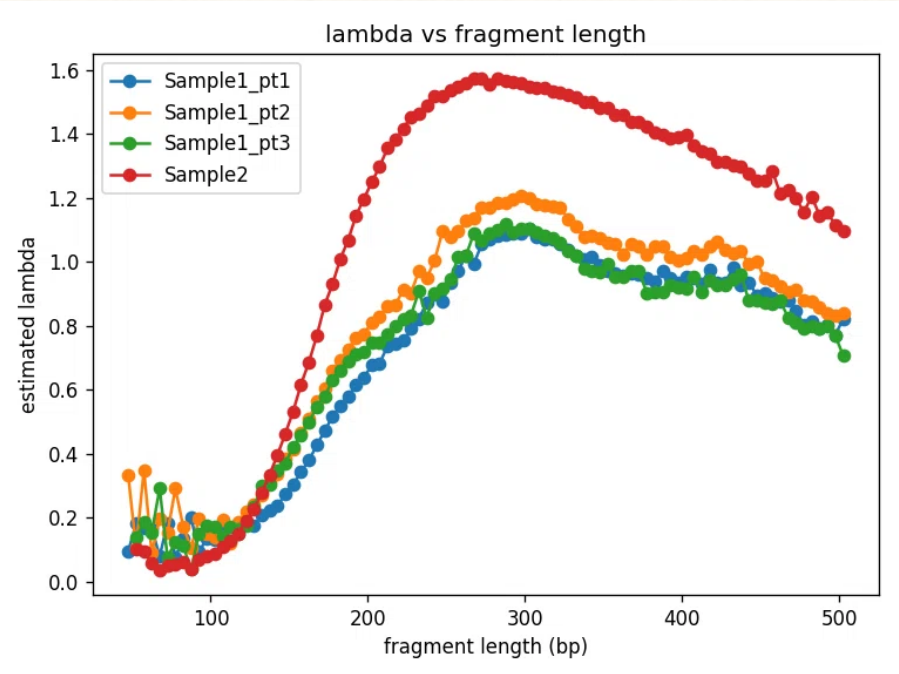
Question 3

a)



estimating λ in each 5bp bin shows that λ is not constant. it's near zero for fragments under 100bp, rises between 150-300, peaks around 280 - 300 bp, then declines toward 500.
λ varies by more than x1e across length
So the assumption that's λ constant is false.

b)



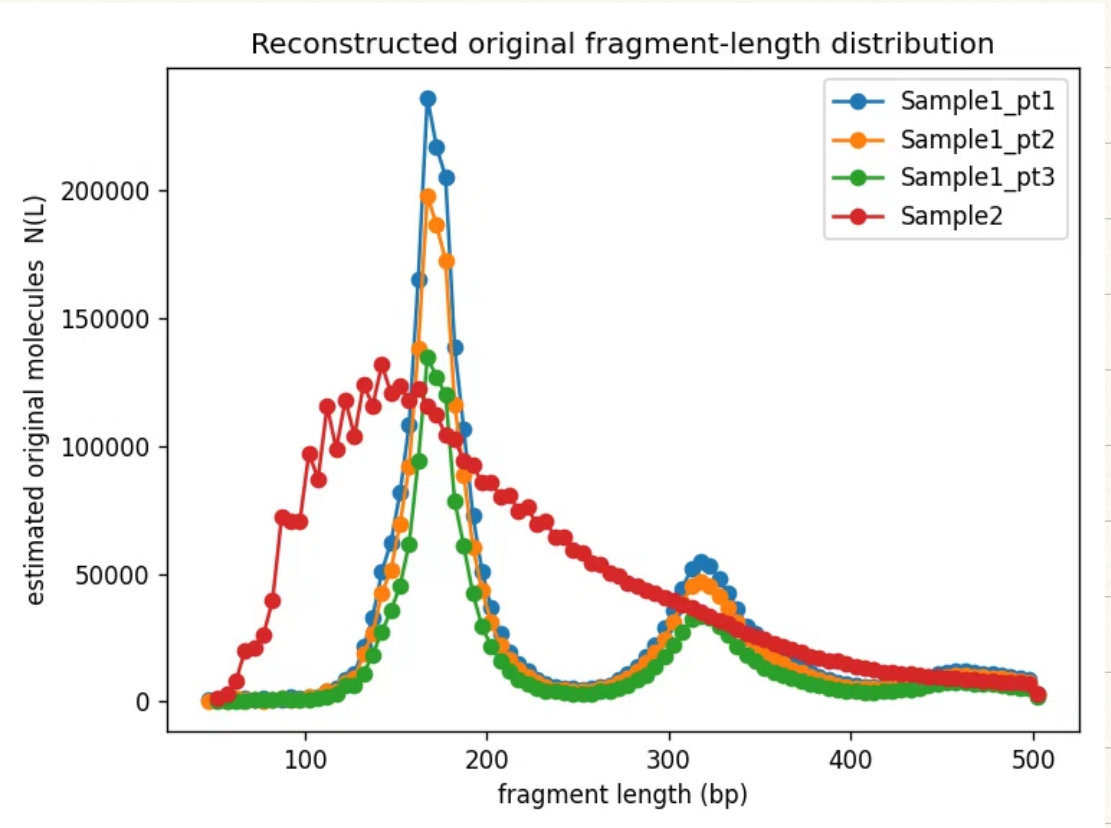
the 3 samples produce almost identical curves in both shape and height, so the behave as a constant process.
Sample 2 has the same shape but higher λ,
=> deeper amplification but same length bias.

to test is there's a difference between 2 curves, I'd put confidance interval on each per-bin λ and check for overlap, or run likelihood ratio test of shared λ model vs. per file λ model

c) λ is per molecule rate, its shape vs length is the amplification / sequencing bias itself, mid length fragments (300bp) duplicate most effeciently while short and long ones are under sampled (likely from size selection and weaker

PCR/ clustering at the extremes).
So complexity isn't uniform across length, sample 2 sits higher but shares the same shape, so the bias mechanism is the same.

d)



I recover the original distribution by correcting each bin for unsampled molecules:

$$N(L) = \frac{M_u(L)}{1 - e^{-2cL}}.$$

this gives a main peak near 170 bp and 2nd one near 320 bp, consistent with nucleosome laddering. the 3 sample1 parts match, sample2 is broader and a little shorter.

the correction matter the most at the short and long ends, where sampling was least complete, revealing the true insert size structure of the starting material.